hydrocarbons, emitted from vehicles, factories and fossil fuel burning. These react with sunlight to produce smog, a thick fog or haze. There is one more component to this mix that is not always visible—particulate matter (PM). Chemicals, soil, smoke, dust or allergens constitute different types of PM. Varying in size, shape and chemical composition, they are defined by their diameter for air quality regulatory purposes.

Pollutants that are not PM are perhaps the lesser evil, though not if you ask young children, the elderly, those with weakened immunity or respiratory ailments. They irritate the eyes and harm the lungs. PM, on the other hand, can affect even healthy individuals. The tiniest of these particles, PM2.5, is 30 times thinner than human hair and thus invisible to the naked eye. It penetrates the lungs. Burning gasoline, oil, diesel fuel or wood produces much of the PM2.5 pollution, whose levels are dangerously high in Delhi NCR. PM10 is also harmful when its level exceeds the safe limit. While some amount of PM10 comes from the same sources as PM2.5, it also includes dust from construction sites, landfills and agricultural activities, wildfires and waste-burning, industrial sources, wind-blown dust from open lands, pollen and fragments of bacteria.

This year, Delhi saw 11 severe air days (with AQI over 390) in the month of November, far more than the three recorded last year for the same month. On November 2, the Anand Vihar bus station in Delhi registered a toxic and poisonous AQI of 999. However, while the national capital gets all the bad press, the November 26 data from the Central Pollution Control Board (CPCB) showed AQI levels exceeding the acceptable limit of 100 in all major cities. It was 109 in Mumbai and 168 in Kolkata. Only Bengaluru at 57 and Chennai at 51 had air fit to breathe. And if you thought you could escape the bad air in Delhi and spend some time in Dehradun, Chandigarh, Gwalior, or Jaipur, their AQIs were 92, 204, 227 and 230, respectively. "It is not an isolated problem," says Dr Nangia. "Nearly every city in the country has high AQI. The WHO recommends an AQI of no more than 50. In India, we take 100 as safe. But most cities can hardly meet even that mark. In Delhi, the AQI is under 100 perhaps only on a handful of days."



"THE WHO RECOMMENDS AN AQI OF UNDER 50. IN INDIA, WE TAKE 100 AS SAFE. BUT MOST CITIES IN INDIA HARDLY MEET EVEN THAT MARK"

DR VIVEK NANGIA, *Principal director, head, pulmonology, Max Super Specialty Hospital, Saket, Delhi*

And given that 67 per cent of our population lives in areas that exceed India's own national air quality standard of 40 micrograms per cubic metre (far in excess of the WHO guideline of PM2.5 concentration no higher than 5 μg/m3), what exactly are we dealing with?

THE EFFECTS OF POLLUTION

PM enters our lungs and is treated by our bodies as a foreign toxin. "Your body reacts to foreign bodies in the same way—whether it is a virus or PM. There is inflammation," says Dr Khanna. Internal inflammation is, of course, invisible, but you can see it on your skin. "It is the red patch that forms around a sore, or a sun burn. Our skin also gets inflamed as an immune response. The same thing happens in our airways when PM enters it," says Dr Rajesh Chawla, a pulmonologist at the Apollo Hospitals, Delhi. It is this inflammation that is linked with a variety of pulmonary diseases, and COPD in particular. "In India, we get COPD from pollution, not smoking, which is otherwise the most common cause globally."

And things are progressively getting worse. "Normally," says Dr Khanna, "we see people with respiratory symptoms come into hospital after Diwali. This year, it has started two weeks prior." Inflamed airways don't just cause disease and discomfort but also prevent the body from recovering from illness. This is perhaps why

THE MOST POLLUTED CITIES

DELHI

Vehicular and industrial emissions keep the national capital's air quality toxic round the year. It has 643 motorised vehicles per 1,000 people. Ahead of winter, stubble burning in Punjab and Haryana worsens the pollution

MUMBAI

Mumbai has the advantage of the sea breeze that disperses the toxic air. But over time, infra projects and construction have led to wind stagnation and dust pollution. Around 6,000 projects have been warned of action if they don't fall in line

KOLKATA

Industrial pollution and elderly vehicles are the chief causes of pollution in the eastern metropolis. Govt estimates suggest there are over 2 mn 15-plus years old motor vehicles plying in the city